A Comparative Semantic Analysis Between Asset-Based and Data-Based Blockchain Interoperability Protocols Using Process Algebra

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Chapter 1

Introduction

Now, far from their original design for cryptocurrencies, blockchain has found its way to incorporate itself into other domains [99] [55] [92] [96]. The main feature that led to this widespread adoption today is its decentralised data storage [54]. Blockchain is a type of ledger that instead of recording its transactions on one medium or it being controlled by a singular entity, its transactions are recorded in several blocks of data storages that are chained from each other and controlled using consensus. Furthermore, once a transaction has been recorded, it becomes unchangeable [**?**]. Aside from decentralisation, it also posseses several benefits..

FundChain is a blockchain implementation built by World Bank for monitoring their use of funds. Through this, they are able to track their disbursed funds from the governemnt down to PIUs, contractors, or to their end-beneficiries [2].

asset transfer had more focus [15] [117]. interoperability is the ability of two or more software componenets to cooperate despite differences in language, interface, and execution platform [15]. asset vs data transfer [15].

Hypothetesise that there will be more effort in building data based solutions even for exchanging assets [15]. This raises the concern more into ensuring that the semantics of the two will not be conflated.

Semantic Interoperability: The capacity to understand and make meaningful use of the data transmitted across many blockchains [92].

blockchain interoperability means connecting multiple blockchains to access information and to

act on the information[55].

In order to create meaningful exchange between blockchains, agreement on the semantic level is necessary[99]. This layer of blockchain interoperability receives little research attention [99]. The semantic layer is responsible for interpreting information format and its meaning.

Addressing these silos is now viewed as an existential requirement for the crypto economy. The success of the multi-chain future depends on the reliable connectivity between these fragmented hubs. This fragmentation has motivated extensive research and development in blockchain interoperability.

Blockchain interoperability in is the ability of distinct and independent blockchains to seamlessly exchange digital assets, data, and arbitrary messages [117] [18] [97]. This is essential if we want the blockchain ecosystems to be a cohesive network. This will prevent the fragmentation of liquidity and utility across the decentralized landscape [117]. Through this, complex applications can now function across multiple blockchains [16]. Early visions for this connected future were established by pivotal whitepapers, including Vitalik Buterin's (2016) conceptualizations for cross-chain connectivity [18] [29]. The Polkadot network, envisioned by Gavin Wood (2016), introduced a heterogeneous multi-chain framework where specialized "parachains" achieve interoperability via a central relay chain

[51] [6]. Similarly, the Cosmos network, outlined by Kwon and Buchman (2016), proposed an "internet of blockchains" architecture using a hub-and-zone topology to facilitate trust-minimized communication through the Inter-Blockchain Communication (IBC) protocol [51] . These early frameworks established the fundamental strategies ranging from lock-and-mint mechanisms to light-client verification—that contemporary bridge protocols utilize today [97].

From a functional standpoint, surveys categorize bridges by whether they support "asset transfer," "data transfer," or both [97]. However, it is critical to note that these prevailing classifications are primarily feature-based. That is, they focus on the operational mechanisms of message passing and tokenization rather than the underlying semantics or the formal correctness of the data being exchanged [51].

The most common significant risk in decentralised systems like decentralised finance or DeFi ecosystem is in the security. This is evident primarily in the cross-chain bridges. Security researchers highlight that they are particularly vulnerable because they often store vast amounts of collateral that back wrapped assets on secondary chains [104] [18]. According to Chainalysis, over $2 billion in cryptocurrency was stolen across 13 separate bridge hacks in 2022 alone. This accounted for approximately 69% of all funds stolen during that year [104]. Another reports from

CertiK and Elliptic further emphasize that the intersection of on-chain smart contract logic and off-chain communication components creates a significantly larger attack surface than that of isolated decentralized applications [48].

1.1 Problem Statement

A recurring pattern can be seen from these systematic bridge failures. These patterns centered on the violation of fundamental protocol invariants [1]. A primary mechanism for these thefts is the unauthorized minting of unbacked tokens. In this mechanism, the attackers exploit the verification vulnerabilities of the system to create assets out of thin air. These assets are created on the destination chain without corresponding collateral on the source chain [81] [27]. This creates imbalance. Other incidents, such as the Nomad and Binance exploits, involved replay attacks or proof manipulation. In these incidents, the bridge was tricked into accepting fraudulent state updates [16] [27]. These real-world failures represent a catastrophic breakdown in the conservation of assets. Here, the mathematical equilibrium that should exist between disparate ledgers was completely severed [1] [29] [73]. These failures reveal that current interoperability approaches often lack clear semantic guarantees about what must always hold true during cross-chain operations.

Semantic compatibility is explicitly identified in the literature as a primary open challenge for blockchain connectivity [16]. This presents a significant barrier to the maturation of the multi-chain ecosystem. Specifically, current frameworks often facilitate the verification and transportation of data between chains without transporting the semantics of its proper use. This leaves a critical gap at the protocol/semantic interface.

How do asset-based and a data-based blockchain interoperability compare in terms of their semantically defined observable behaviour? Specifically, this research aims to answer the following research questions:

1. How can asset-based and data-based blockchain interoperability protocols be formally defined?
2. What observable behaviours emerge form the semantic models?
3. In what ways do these obsservabe behaviours differ or align under a semantic framework?

increasing need for data interop, so we find out the semantic of these crypto based interop, and the semantic of data- based interop, and then find their differences.

1.2 Objectives

In order to address the lack of formal semantic distinction between asset and data interoperability, this research proposes the following objectives:

1.2.1 Main Objective

To develop a formal semantic framework that distinguishes asset from data interoperability in cross-chain systems using state-machine modeling and invariant analysis to provide rigorous safety guarantees for multi-chain interactions.

1.2.2 Specific Objectives

- To formalize asset interoperability semantics by defining state variables, transition functions, and conservation invariants, which are necessary to ensure the mathematical equilibrium of tokens across disparate ledgers
- To model data interoperability semantics by specifying state variables for arbitrary message passing, rules for cross-chain delivery, and integrity properties, which ensure that data remains synchronized and correctly interpreted across heterogeneous environments.
- To conduct a comparative analysis of the invariant structures for asset and data transfer to identify fundamental semantic differences, specifically focusing on how safety, liveness, and fault tolerance are uniquely preserved in each category.
- To demonstrate the framework's utility by mapping existing interoperability protocols and their documented failure modes (such as the Wormhole and Nomad exploits) to the proposed semantic categories to verify their adherence to fundamental protocol invariants.

1.3 Scope and Limitations

This research focuses on the underlying logical properties rather than the implementation details or empirical performance of certain blockchain protocols.

1.3.1 Scope

This research will use state-machine modelling invariant analysis to formalise how information and value move between independent ledgers plus identify and verify critical protocol invariants like conservation of assets and message integrity. This research will analyse established patterns in asset and data interoperability in order to develop a unified model to bridge the gap between protocol-level details and semantic-level interoperability. It will not propose new interoperability protocols nor provide functional bridge implementations.

This research will not benchmark metrics such as transactions per second (TPS), latency, or gas efficiency. It will not analyze the internal mathematical robustness of cryptographic primitives such as secp256k1 and SHA2-256. It will not focus on providing machine-checked proofs in languages like Coq or Isabelle/HOL; but rather on framework definition and manual formalization. It will not consider incentive compatibility as well such as validator reward structures, or the economic cost of attacks. Lastly, legal and compliance frameworks, such as AML/KYC requirements or jurisdictional laws, are outside the scope of this technical analysis,.

1.3.2 Limitations

The proposal models in this research are abstract and minimal by design. They prioritize the "core logic" of state transitions and may not capture every low-level implementation detail of real-world protocols. This research limits the analysis to a selected set of representative protocols like Ethereum, Cosmos and Wormhole. It is not intended to be an exhaustive survey of all existing bridging solutions. This research does not implement testing on live testnets or mainnets; the verification remains at the formal specification level. Lastly, the findings are primarily theoretical and intended to inform design patterns. The practical deployment may require further adaptation to account for off-chain behaviors and network synchrony.

1.4 Significance

This research fills a critical gap in existing literature by providing the formal semantic distinction between asset and data interoperability. It contributes a foundational abstraction for interoperability research. Thus, it directly address the "understudied aspect" of semantic compatibility identified in major surveys. By defining the protocol/semantic interface, this work enables more precise protocol

classification and comparison than the prevailing feature-based taxonomies. Furthermore, the clarity that this framework provides by specifying the invariant requirements that protocols must uphold, future formal verification will be easier and robust. Thus also helps provides rigorous basis for automated reasoning in distribute systems.

From a practical standpoint, this framework also helps protocol designers and architects. This research will help them understand the distinct semantic requirements of different interoperability models. It will clarify the technical necessity for conservation guarantees in asset transfers while establishing that data transfers must prioritize integrity and schema alignment. By formalizing these requirements, the research aims to prevent design errors that have historically led to multi-million dollar bridge vulnerabilities, such as the $320 million Wormhole exploit and the $190 million Nomad hack. Additionally, it provides essential conceptual clarity for security analysis and auditing, moving the industry toward standardized security best practices and safer cross-chain communication protocols.

Chapter 2

Review of Related Literature

In the current landscape of blockchain technology, the emergence of heterogeneous networks has revealed the term "information isolated isalands" or "value silos" [111] [20] [121]. Although individual blockchains are STRONG/GREAT as independent ledgers, the lack of communication between them limits their potential widespread adoption [47] [32] [69]. Blockchain interoperability then is not just a feature; but a critical interaction between distributed systems [111] [63] [69].

It is vital to understand that every blockchain acts as an independent state machine with its own concensus rules and encryption standards [20] [17]. Since there are no shared state or global clock between these different blockchains, their interactions must happen asynchronously; they must instead use message passing and state synchronisation [111] [115]. In other words, blockchain interoperability is a process wherein one source chain is trying to change the state of a target chain by means of delivering information or assets under certain securoty assumptions [111] [112] [16].

Most of study in the literature focus on the blockchain interoperability architecture. That is, whether it uses relays, sidechains, or notaries [55] [25] [62] [30]. Nevertheless, recent studies have put emphasis that architecture only gives stucture and real security and effectiveness of interoperability depends on the behaviour of the system [67] [66] [69]. This behaviour dictates how systems guarantee ACID (atomicity, consistency, isolation, durability) properties in an asynchronous network [111] [20] [112] [63].

This means that the sequencing of state transitions, timing of cyrptographic proofs, and reactions to failure or timeout reveals the system's behaviour [84] [33] [37] [115].The analysis of behaviours,

instead of the static design, gives deeper understanding on how different components (such as smart contracts and relayers) interact during different processe. That is so to achieve semantic interoperability where the meaning and context of data or asset from the source chain gets accurately interpreted and valid to the target chain [14] [99] [63].

2.1 Approaches to Blockchain Interoperability

Surveying the literature, it is noticeable that the classifications of interoperability reflects the underlying logic on how to trust and treat information between blockchains [111] [86] [113]. Although categorical labels like notary schemes, sidechains/relays, and hash-locking are usually used, there is a deeper abstraction that surfaces in recent literature. That is, the perspective of looking at blockchain interoperability as a multi-layered problem derived from the OSI model of the internet [47] [37].

The common abstraction from most of the sources is the **source-target model**. In this model, an operation from the source chain needs to trigger a corresponding state change to the target chain [37] [32]. Nonetheless, here comes the critical difference on how this interaction gets seen:

Interoperability as Transfer: In this perspective, the logic focuses on migration or transfer of ownership. Every transaction must follow all-or-nothing atomicity in order to avoid double-spending [111] [65] [37]. Here belongs asset-based protocols that ensure that the value will not vanish or duplicated between silos [14] [113].

Interoperability as Communication: Here, the system gets treated as a **message-oriented service** [111] [55] [49]. The goal is not to transfer the content of the source chain but to invoke a functionality in the target chain using arbitrary data [86] [17]. This is the logic of the data-based protocols like Arbitrary Message Passing (AMP) [11].

It is critical to make the distinction between asset-based and data-based interoperability because they usually remain implicit in a number of older literature. For instance, in the classical SoK (Systematisation of Knowledge) of Buterin, the focus was always in value exchange or value islands [111] [3] [86]. However, in the newer frameworks, there is an explicit division in the domain between **Asset Interoperability** (migration/exchange) and **Data Interoperability** (cross-chain calls/integration) [37] [32] [63]. This gap is important as a number of current models are designed for assets but gets applied to data or vice-versa without taking each different semantic requirements

into consideration [14] [63].

This differeng individual trust boundary and security assumptions of each approaches contributes to the fragmented landscape of blockchain interoperability[14]. Although there is an attemot to have a Generic Interoperability Paradigm comprising of Setup, Commit, Verify, and Abort phases, the literature has not fully explained how these phases differ in their **observable behaviour** when assets are passed isnstead of arbitrary data [37] [63]. This forms the need of a **comparative semantic analysis** to see if these models are enough to encompass the behaviour of both asset and data exchanges [63].

2.2 Asset-Based Interoperability

The literature treats asset-based interoperability not solely as a tool to transfer currency but also as a **closed system of value synchronisation** [86] [32]. The fundamental logic of these protocols is the preservation of the INTEGRITY of assets while they are moving between different ledgers [65] [86] [14].

2.2.1 System Perspective

As a system, asset-based interoperability comprises the following interdependent components: **Chains (Source and Target):** These serve as state machines that record ownership [111] [37] [32]. **Smart Contracts (Custodian vs. Debt Issuer):** In the source chain, there is usually a custodian contract (vault) that locks asset; while in the target chain, there is a debt issuer (money printer) that makes synthetic representation or wrapped tokens [82] [100]. **Users and Intermediaries:** Included here is the initiator (user), relayers that transfer proofs, and attesters that verify if the lock event is valid before doing the minting [82] [67] [14].

This shows that asset-based interoperability is a content-aware system. Meaning, it is necessary that the protocol knows the specific semantics of asset (e.g. ERC-20 standard) in order to ensure that the minted token is semantically equal to the original asset [67] [86] [37].

2.2.2 Behavioural Patterns: Logic of Ordering

The observable behaviour of asset-based protocols follow a strict sequential execution flow. Based on existing formal models (such as Event-B analysis), the usual sequence follows: **Lock/Burn**

(Source): Immobilisation of the original value [82] [67] [37]. **Verification:** Passing the Merkle proof or SPV (Simplified Payment Verification) proceeding to the target chain [111] [70] [20] [37]. **Mint/Claim:** Producing of an equal value at the target chain [67] [98] [63].

This shows that there is happens-before relationship. The mint transaction at the target chain has causal dependency on the lock transaction at the source chain [67] . If this sequence is not followed, there is a risk of atomicity failure where there is a possibility of minting assets that are not locked; resulting to double-spending [65] [14] [37] [62].

2.2.3 Invariants and Safety Guarantees

The major invariant that safeguards this system is the Global Supply Invariant. This refers to maintaning a constant value to the overall count of asset units in all networks [86] [62] [20].

It is critical that the safety (no bad behaviour or duplication) of asset-based protocols gets prioritised over the liveness (fast termination) [67] [20]. Because of this, the observable behaviour has long confirmation periods (to wait the finality from the source chain) and competition periods (for fraud proofs) [37] [111] [112] [72]. These delays are not just performance issue but a behavioural necessity in order to protect the atomicity of the system [33] [70] [32].

Nevertheless, this behaviour results to capital inefficiency [100] [78] [63]. That is because of the locking of assets, the liquidity gets trapped in the contracts, that serves as a big limitation that is not usually seen in data-based approaches [100] [37] [11]. This creates the gap in the literature where even though the safety of asset transfers can be formally modelled, there is insufficient comparison if this strict ordering is also needed for arbitrary data exchange [67] [66].

2.3 Data-Based Interoperability

Unlike asset-based interoperability where it is illustrated as a mechanism of changing ownership, the literature describes data-based interoperability as a decentralised communication system [70] [20] [55]. If asset-based focuses on liquidity, data-based on the other hand focuses on the arbitrary message passimg (AMP). This is a mechanism where the system acts as a transport layer for whatever type of information [36] [11].

2.3.1 System Perspective: The TCP/IP of Blockchains

Being a communication system, its architecture comprises of the following interdependent components: **Chains and Modules:** Every blockchain hosts modules or handlers (smart contracts). They serve as entry and exit points of data [36] [33]. **Relayers and Connectors:** Since blockchains are isolated islands and incapable of sending data natively, they need off-chain relayers (such as Hermes) that scans blocks for pending messages and transfer them to the target chain [36] [33]. **Channels and Connections:** These are the stateful abstractions that maintains logical link between two chains that gives ordering and authentication to packets [20] [115].

The literature sees the system as a payload-agnostic service. This shows that the protocols do not care whether what is inside the packets is a token transfer or a complex smart contract call. Its purpose is to ensure that the opaque bytes are delivered with integrity [55] [36] [33] [20].

2.3.2 Behavioural Patterns: Handshake and Asynchrony

The observable behaviour of data-based interoperability is guided by handshake protocols and asynchronous execution [70] [20] [116]. **Handshake Process:** This is unlike the simple lock-and-mint; data-based systems follows a strict 4-step handshake (OpenInit -> OpenTry -> OpenAck -> OpenConfirm) in order to establish a connection [20] [116]. This behaviour is needed to ensure that both chains are ready to accept data under the same security assumptions [116]. **Asynchronous Message Passing:** The interaction here is non-blocking. The source chain delivers a message and can proceed to other operations while the packet is in-flight [70] [20]. The success of the interaction depends on Acknowledgements (ACK) or Timeouts. They are observable events that dictates if an operation is deemed Delivered or Expired [33] [116].

2.3.3 Comparison to Asset-Based Interoperability

The fundamental difference between the two is rooted on their interaction logic: **State-Dependent vs. Instruction-Dependent:** Unlike the asset-based that depends on state provenance (it needs the verification of the lock state before minting), data-based behaviour is instruction-dependent [114] [37]. Its aim is to trigger a functionality in the target system that will work under its own rules [111] [17]. **Atomicity vs. Eventual Delivery:** In asset transfers, atomicity is commonly all-or-nothing under a capital constraint [86] [14]. Meanwhile, in data exchange, it prioritises liveness and delivery

guarantees. It means that every message should have a definite result (accepted or timeout) inside the asynchronous network [20] [86] [116].

The behaviour here depends on the reliabilit of the transport layer. Nonetheless, the critique in the current literature is the dependency of these systems to untrusted layers [116] [36]. Formal analyses (such as the TLA+) shows that even the standardized protocols can have trapped packets or incomplete handshakes if the relayers' behaviour cannot be properly modelled [115]. This pushed the need for deeper comparative semantic analysis to see how it is possible to converge the safety of assets and the liveness of data in one behavioural model.

2.4 Formal Methods to Blockchain Interoperability

In the study of interoperability protocols, simple architecture diagrams alone are insufficient to guarantee the security and correctness of systems. Formal Methods surfaces in the literature as critical tools to avoid design flaws that can lead to large financial losses [115] [66]. In surverying the literature, it is noticeable that each formal method has its own modelling philosophy that dictates what aspects of behaviour are being emphasized.

2.4.1 Modelling Philosophies and Methodogies

From what has been gathered in the literature, formalisations done on blockchain interoperability is divided into four major philosophies: **Event-B (State + Invariants):** This approach in focused on modelling the system as a discrete transition system [66]. The strength of this is in verifying the safety properties of the system using mathematical proofs and invariants. For example, this is used to ensure that a token cannot be minted if it is not yet officially locked on the source chain [67]. Nevertheless, this philosophy has its own limitation in capturing the liveness (or the guarantee that something will happen at the end), and is usually struggles in modelling asynchronous timing issues [67] [66]. **TLA+ (Traces + Temporal Behaviour):** In data-based protocols like IBC, TLA+ is the dominating method [116] [67]. Its philosophy is rooted in Linear Temporal Logic where the behaviour is seen as a sequence of states or traces. This is effective in detecting bugs in asynchronous communication like trapped packets or incomplete handshakes [116]. However, because of its high level of abstraction, it usually disregards the internal logic of smart contracts [66] [116]. **Abstract State Machine or ASM (Abstrac Execution):** The philosophy is to

abstract the whole blockchain as a deterministic transition function [70]. Its focus is the operational semantics of the remote calls and how they maintain atomicity in multi-chain transactions. This focuses mmore on the execution flow more than the static invariants of assets. **CSP# and pi-calculus (Interaction + Concurrency):** These methods focus on process algebra [110]. Its philosophy is in the treatment of interoperability as a series of inter-process communications. This is useful in modelling interactive protocols like atomic swaps where it is needed that two untrusted parties simultaneously do an action.

2.5 Synthesis and Research Gap

In surveying the literature, a clear pattern surfaces: although the improvement in the technical solutions to blokchain interoperability is growing fast, its theoretical and formal foundation still remains slow and fragmented. The following discussions lay outs the critical gaps that reasons the need for this research.

2.5.1 Fragmentaion in Knowledge and Technology

Every study in the current literature focuses on particular silo or isolated implementations. It can be observed that the current research landscape is divided into walled gardens where one group of the literature is exclusively discussing asset-specific bridges while the others solely focus on messaging-agnostic hubs. For this reason, the understanding of interoperability is not being hollistic. This fragmentation is not only technical but also conceptual; there is no unity in the literature on what really is the standard interaction process to be followed by all protocols.

2.5.2 Lack of Unified Abstraction

There is no unified way to represent the semantics of cross-chain interactions using a framework that is not dependent on codes. Most of the gathered studies give ad hoc architectures or intentionally made for one particular use case. This shows that the literature currently has insufficient behavioural primitives. These are guiding action units like locking, proving, or emitting that can be used to describe any protocol whether it is an asset-based or data-based. Without this abstraction, developers and researchers will struggle in verifying the security of systems when different protocols gets interconnected.

2.5.3 Lack in Direct Comparison of Behaviours

There is direct comparison done on the behaviour of asset-based and data-based protocols inside one formal semantic framework. Although there are surveys that list features such as whether they are decentralised or have notary, they do not survey the fundamental difference in terms of interaction logic. Asset-based protocols are described as state-dependent (it needs lock before minting), while data-based are instruction-depended (triggering functions) but the implication of this difference in their observable behaviour is not fomally written. Meaning, although their goal is known, their difference in their execution traces cannot be formally shown.

2.5.4 Methodological Gap in Formal Modelling

Most of the current formal models are state-based, not interaction-based. The literature is currently limited to one dichotomy: The usage of Event-B for verifying static invariants and safety of assets and the usage of TLA+ to survey liveness and traces of data packets. This gap is having no model that can unify the low-level state transitions (ownership changes) and high-level messaging flow (function calls) in one language. This does not directly analyses the interaction of value and data as connected manifestation of system behaviour.

2.5.5 Conclusion and Research Direction

In conclusion, there is a need to have an approach that will abstract these protocols as formal systems of interaction instead solely as implementations. The current gap in the literature is not just only a topic of fastness or throughput, but the lack of deep semantic understanding on how blockchain interoperability operates on the behavioural level. One possible way is the usage of a comparative semantic analysis that focus on the creation of formal model for asset and data transfers. This research aims to fill the aforementioned gaps by means of identifying the observable behaviours that will give clarity to the similarities and differences of the two paradigm. This will serve as a guideline for making more robust and standardized interoperability frameworks in the future.

Table 2.1: Matrix

Citation	Type	Interoperability Type	Formalisation	Limitations
[100]	Survey/Protocol Design	Asset		

[82]	Survey/SoK	Hybrid (bridges)	-	Lacks formal system model
[25]				
[11]				
[63]				
[36]				
[52]				
[37]				
[3]				
[47]				
[56]				
[67]				
[20]				
[70]				
[65]				
[66]				
[32]				
[99]				
[78]				
[61]				
[110]				
[116]				
[10]				
[30]				
[62]				
[113]				
[14]				
[121]				
[46]				
[55]				
[33]				

[72]

[114]

[98]

[49]

[84]

[86]

[69]

[17]

[111]

Chapter 3

Theoretical Framework

3.1 Systems and System Behaviour

A **system** is formally composed of two major aspects: process and data. In this framework, processes serve as control mechanisms that runs and manipulates data. Meanwhile, data remains static and passively comprises the system [40]. The composition of these elements formsa structure that can perform operations according to specified rules [88].

The **behaviour** of a system refers to the sum of its ability to interact or communicate. A behaviour is **observable** because it can be seen from its outside operations; particularly, the interaction of the inputs it accepts and the outputs it produces [77]. Every elementary step or operation that happens in the system is called **transition** that changes the system from its current configuration to another state [88].

A **Blockchain** is an example of a system that can be treated as a transaction-based **state machine** [119]. In this view, the blockchain is not just a storage of information but a dynamic mechanism where transactions serve as inputs that result to transitions of the distributed ledger's state. Every succesful transaction creates new state that is formally accepted by the whole network. The usage of **formal modelling** is critical in this topic in order to give sufficient mathematical evidence for the security and stability of interoperability systems. Formal modelling removes ambiguity and ensures the correctness in comparing asset-bsed and data-based mechanisms under one theoretical framework [38] [40].

3.2 Semantic Modelling Using Labelled Transition Systems (LTS)

Labelled Transition System (LTS) is a fundamental mathematical structure used to model computations and dynamic behaviour of systems [107]. It represents systems as states and transitions labelled by actions.

However, in order to capture the complex behaviour of blockchain interoperability, the system is modelled as a **Probabilistic Lablled Transition System (pLTS)**. It is composed of quadruple

$$(S, Act, P, \rightarrow)$$

. In this framework, the transition relation ($\rightarrow$) does not only connects states but also refers to a probability distribution (Δ) above the group of states (S). This probability distribution is crucial in modelling nondeterministic environments of consensus protocols and frame collisions [108].

The **observable behaviour** of the system is composed of the **traces**. They are the series of actions from the starting state [43]. This formalism allows the abstraction from the inside implementation to the outside communication [88]. These traces serve as the guidelines to analyse whether a protocol follows its intended semantic interoperability.

It is important to notice when two systems with differing structure representation or graph can share the same behaviour [13]. This concept is called **behavioural equivalence** wherein the states of different LTSs are treated as equal if they exhibit the same pattern of transitions such as what the theory of bisimulation describes [13] [43]. This abstraction allows the inquiry of comparison of protocols for asset-based and data-based interoperability in blockchains without limiting particular details of their internal implementation [88].

Nevertheless, the simple usage of state transition graphs can lead to the problem of state explosion that makes formal verification difficult [105]. For this reason, there is a need to a more expressive framework that allows algebraic manipulation and composition of processes.

3.3 Process Algebra for System Composition and Communication

Process Algebra serve as the main theoretical instrument of this research particularly the principles of communication, composition, and abstraction. The research will use it to formally analyse and manipulate operations of blockchains. This is illustrated as the study of concurrent systems or communicating systems inside an axiomatic and algebraic framework [19] [40]. Instead of just using visual diagrams, processes are embodies as formal *process terms* using a mathematical language that allows linear notation and mechanised manipulation [40].

3.3.1 Atomic Actions and Alphabet

System behaviour is constructed from a formally defined alphabet of atomic actions. Let:

$$D : \text{set of data}$$

$$P : \text{set of communication ports}$$

Atomic actions are defined as:

$$Send : s_p(d)$$

$$Read : r_p(d)$$

where $d \in D$ and $p \in P$ [106].

3.3.2 Communication and Handshaking

Communication between processes is governed by a communication function:

$$\gamma(s_p(d), r_p(d)) = c_p(d)$$

This formalises the handshaking paradigm where communication happens only through synchronisation between matching send and read actions. Isolated actions are disallowed to ensure correctness in system interaction [106].

3.3.3 Core Operators

In this framework, every system's process is composed of atomic actions (actions that has no duration) combined using the following operators: **Sequential Composition** ($x \cdot y$)**:** x is perfomed first before y. **Alternative Composition** ($x + y$)**:** symbolises the nondeterministic choice between x and y operations. **Parallel Composition** ($x||y$)**:** symbolises the simultaneous execution of actions x and y.

A fundamental feature of process algebra is its **compositionality** wherein a large system can be treated as combined smaller and simpler parts [40]. Using the **parallel operator** (||), individual blockchain network or protocol components can be modelled separately and then regouped to see the whole behaviour of the inter-operational system [40]. This modular technique allows the analysis of the connection between asset-based and data-based protocols as a series of concurrent processes [40].

3.3.4 Encapsulation and Abstraction

To control system visibility: **Encapsulation** (∂_H) blocks unmatched internal actions. It is used in order to ensure these internal actions do not happen alone but as a part if a synchronized communciation that causes transition to the state of the whole composition. **Abstraction** (τ_I) hides internal communications as silent steps (τ) [40].

3.3.5 Timed and Reliable Behaviour

Since interoperability protocols operate in asynchronous environments, timing and reliability must be incorporated. The framework adopts the principles of **Positive Acknowledgement with Retransmission (PAR):** Successful delivery $\rightarrow$ acknowledgement (ac) Failure or delay $\rightarrow$ timeout (to) Recovery $\rightarrow$ retransmission To regulate behaviour, a priority operator (θ) is introduced. It assigns lower priority to timeout actions than successful communications to prevent premature retransmissions.

Also, process algebra alone may not be sufficient in modelling interoperability mechanisms like HTLC. Because these mechanisms depend on timeouts, the formalism needs to be a Timed Process Algebra where in each process, there is a local variable *now* to keep track the passing of discrete time [108].

3.3.6 Operation Semantics

The meaning of process expressions is formally defined using **Structural Operational Semantics (SOS)**. Transition rules specify how processes evolve. This ensures that every behaviour is derivable and mathematically well-defined [106].

3.3.7 Behavioural Equivalence and Verification

In order to identify whether an implementation of a protocol matches to its expected behaviour, the concept of **behavioural equivalence** is used; particularly, the **branching bisimulation** ($\leftrightarrow_{rb}$) [41] [40]. Branching simulation is needed because it maintains the branching structure of processes that is crucial in identifying deadlocks which can result to loss of value or message [41]. In order to prove this bisimulation between two different ledgers, **Cones and Foci methodology** must be used where every state in an implementation is mapped to a particular focus point at a specification that shows similar external behaviour [41].

3.4 Interoperability as System Composition

In process theory, interoperability can be formally described as a composition of two or more systems that work together to achieve a particular goal [58]. Under the framework of Process Algebra, this communication is symbolised by the parallel composition ($P||Q$), wherein every blockchain network is treated as separate process that runs concurrently and independently from the other networks but with the ability to communicate to each other.

The communication inside this composition is run by the mechanism of **synchronisation** [77]. By means of a formal **communication function (γ)**, the individual actions of every blokchain gets to communicate. For example, the action that delivers the message (*send*) from a system needs to match the action of receiving (*read*) at the other system in order to complete a succefull communication.

The behaviour of the formed composition is treated as observable from the perspective of an outsider viewer [77]. Using the **abstraction operator (τ_I),** the complex details of the internal communication between blockchains are hidden or made as *silent steps (τ),* so that what only remains are the external input-output relation of the whole inter-operational system [40]. The

resulting traces or series observable actions serve as the evidence if the operation of the composed systems remain consistent to the original semantic specification.

This framework of formal composition gives way to formally analyse and compare the two ways of blockchain communications: the asset-based and data-based interoperability.

3.5 Distinction Between Asset-Based and Data-Based Interoperability

In order to analyse the semantic behaviour of interoperability protocols, it is important to identify the difference between the two major category of interation in blockchain: asset-based and data-based interoperability.

Asset-based interoperability is formally illustrated as a mechanism to transfer ownership or value between different blockchain network. Under this model, a digital asset, like cryptocurrency, is formally burned or locked from its source chain in order to create an equal value at its destination chain. The primary goal here is the maintaining of the linear logic of the value wherein one asset cannot exit simultaneously in two different systems.

On the other hand, data-based interoperability is focused on the exhange of information or arbitrary data without needing to have a definite value. This type of interoperability allows a blockchain to read or verify the state of other blockchain, which is usually used to trigger **smart contract invocation**. Instead of transferring ownership, its focus is the delivery of messages from an external source towards an on-chain environment which is usually run by mechanisms like Oracle.

This difference leads to different behavioural implication for each protocol. In asset-based interoperability, the behaviour of the system is critcally tied to atomicity or the guarantee that the whole transaction between chains will be all successful or failure in order to avoid losing of value. In data-based interoperability, the behaviour is focused on semantic consistency and data integrity, wherein the goal is to make sure that the message received by the destination chain is authentic and true to its representation in the original state from where it came from.

The understanding of this difference will lead way to the creation of a formal model that will analyse their transitions.

Chapter 4

Methodology

The research methodology comprises multi-stage pipeline. It will systematically process interoperability protocols from the technical specifications towards the formal mathematical verification. Every stage in this methodology is treated as sequential and dependent; the success of every following stage directly follows from the output of the previous stage. This ensures the logical connection of the whole process.

4.1 Formal Specification of Interoperability Protocols

The goal of this stage is to create a strict representation of representative protocols using algebraic language. This formalisation is important to remove details of their implementation that have no connection to the semantic behaviour. This allows comparison between asset-based and data-based mechanism.

4.1.1 Select Representative Protocols

Protocols that have wide adoptation and formal documentation will be used. For asset-based, Hashed Time-Lock Contracts (HTLC) is considered because it serves as the de-facto guideline for atomic swap. For data-based, Oracle-based Message Relay is considered because of its fundamental paper in delivering external data towards blockchains.

4.1.2 Identify Actions

Identify the elementary operations that makes up each protocol: **Asset-Based** (A_{asset})**:** *lock* (locking of asset), *unlock* (moving asset out using secret), *refund* (returning of token after a timeout). **Data-based** (A_{data})**:** *request* (asking of data), *verify* (confirmation of data integrity), *relay* (delivery of proven message to destination chain). Let:

$$D : \text{set of data (assets or messages)}$$

$$P : \text{set of ports (e.g., source chain, relayer, destination chain)}$$

Then:

$$A = s_p(d), r_p(d) | d \in D, p \in P$$

Protocol specific actions are mapped into this structure: Asset-based:

$$lock \rightarrow s_p(d)$$

$$unlock \rightarrow c_p(d) \text{ (succesful transfer)}$$

$$refund \rightarrow s_p(d) \text{ (reverse flow)}$$

Data-based:

$$request \rightarrow s_p(d)$$

$$verify \rightarrow \text{ internal communication}$$

$$relay \rightarrow c_p(d)$$

4.1.3 Encode Process

Form the logical protocol using the following operators: **Sequential ($\cdot$):** To signify strict sequential process. **Alternative ($+$):** To model branching of decision. **Parallel ($||$):** To model interaction between two different blockchain ledger. System components (e.g., sender, channel, receiver) are modelled as separate processes and composed as:

$$S||K||R||L$$

4.1.4 Encapsulation and Abstraction

To control system visibility: **Encapsulation set:**

$$H = s_p(d), r_p(d)|p \text{ is internal}$$

Ensures that unmatched actions are blocked. **Abstraction set:**

$$I = c_p(d)|p \text{ is internal}$$

Internal communications are hidden as silent steps (τ).

4.1.5 Final System Representation

The complete system can be expressed as:

$$\tau_I(\partial_H(S||K||R||L))$$

This serves as the implementation model (IMP) that will be used for analysis.

4.2 Semantic Extraction Using Labelled Transition System

The goal of this stage is to give the process terms made from the preceding stage formal interpretation using Labelled Transition System (LTS). Through this, every algebraic term will have its corresponding dynamic model that shows all possible observable behaviour of the system.

4.2.1 Lay out Structural Operational Semantics (SOS)

SOS will be used in order to see the meaning of every operator in the process algebra using inference rules. Every rule is composed of premise and one conclusion:

$$\frac{premises}{conclusion}$$

4.2.2 Construct the LTS Model

For every specific protocol, LTS will be formed as a triple

$$(S, A, \rightarrow)$$

:

States (S)**:** set of states comprising the original process term and all terms that can be achieved from it (reachable states). **Labels** (A)**:** labels of transition that refers to atomic actions that the system executes. **Transition** ($\rightarrow$)**:** relation of transition ($s \xrightarrow{a} s'$) that shows change from one state to the another after the action (a). **pLTS** will be used when probabilisq behavious is included: captures message loss, corruption, and nondeterminism.

4.2.3 Extract Traces at Reachable States

Reflexive States: reflexive transitive closure ($\rightarrow *$) to identify all possible states that can be reached by the protocol from the starting state. **Traces:** get the finite sequences of actions

$$a_1, a_2, ..., a_n$$

that the system executed in order to illustrate the totally of its oberservable behaviour.

4.3 Description of Observable Behaviour

The goal of this stage is to formally identify the oberservable behaviour of the interoperability protocols using the extraction of traces and execution paths from the LTS mdoels. This survey will serve as the basis of comparison of asset-based and data-based mechanims at the noticeable level of their connecition.

4.3.1 Give meaning to Observable

The behaviour of a system is tread as oberservable if this comprises actions that can be seen or experience by an outside observer or adjacent system by means of communication. An observable behaviour is formally defined as an external specification (SPEC) that remains after applying the **abstraction operator** (τ_I) in an implementation (IMP) TWOSIMPLEPROTCOLS. In this methodology, the internal connections between chains are treated as **hidden actions** and are made as silent steps (τ) in order to only see the connection of input and output between the system and its environenment . The rigour of the observation does not come from the absence of τ but on how abstraction lessens the complexity of the internal structure towards a simple SPEC that affirms the correctness of interoperability.

Internal calculations or local changes in the storage that cannot be seen at cross-chain communication will not be included in this survery.

4.3.2 Extract Traces and Execution Paths

Traces: Traces ($Tr(q)$) will be extracted by listing all possible series of oberservable actions that can be execute by the protocol from its origin state (q_0). A trace is a string $\sigma \in A*$ where there persist a sequential transitions towardsa new state. **Execution Path:** In order to survey the structure of the protocol deeper, execution paths or the total flow of the transition ($s \xrightarrow{a} s'$) that shows not only the series of actions but also points where there is nondeterministic choice or branch of decision.

Justify as Basis of Interoperability: The oberservable behaviour is the fundamental basis of interoperability because the communication between two blockchain is being treated as a black box interaction. In a chain's perspective, the internal implementation of the other chain has no corelation; what only matters is the actions being sent and being accepted by it in order to make sure the semantic consistency. In other words, the correctness of interoperability depends on whether the observable traces of the two system matches under the one specified protocol.

4.4 Evaluation Criteria

The goal of this stage is to set the formal guideline that will be used in comparing asset-bsed and data-based protocols. By means of identigying the level of similarity and difference of their behaviour, results of the analysation under one concept of semantic interoperability can be given interpretations.

4.4.1 Similarity, Difference, and Equivalence

Equivalence: Two protocol is equivalent if their process graph shows equal pattern of interaction to their environment. Formally, this is identified as an equivalence class ($|x| = y : x \sim y$) wherein the system belonging to this has one oberservable behaviour. **Similarity:** This refers to having some parts of the trace matching even if the structure of the process is not fully similar. **Difference:** This surfaces when a protocol has branching structure or deadlock behaviour that cannot be found in another even if they have matching strings of action.

4.4.2 Evaluation Criterias

Trace Equivalence ($=_{tr}$): This criteria will be used to survey if the protocols can execute exactly the same series or strings of atomic action. Nonetheless, trace equivalence is considered as coarse because this does not capture the difference from deadlock or the timing of choices. **Bisimulation Equivalence ($\sim$):** This the stricter criteria (finest). Under this criteria, two protocol need to have the same branching structure; meaning, in every step of a protocol, the other should respond to the same action and reach a state that is again equal.

4.4.3 Interpretation of the Results

The presence of bisimulation between asset-based and data-based protocol means that the two has semantic interoperability. That is, the meaning and context of their operation still remains consistent in every transition.

If the protocols are trace equivalent only but not bisimilar, the interpretation will have different level of liveness and safety; for example, one fo them can lead to deadlock at situations where the other one have not experienced yet.

The adherence to this formal criterias ensures that interoperability does not only depend on technical connection but also to mathematical correctness and semantic integrity of the system.

Appendix A

Use of Generative AI

The proponents utilised three generative AI: ChatGPT, Deepseek, NotbeookLM. However, there are no set rules on which to use for specific topics. The proponents use either of the three depending on their convenience.

A.1 Brainstorming

Since generative AI tools like ChatGPT can quickly provide answers, it was used to ask trends that are happening in the field of Blockchain. It was asked to provide recent published papers that were checked by the proponents. It would give several research titles with the authors and links. The proponents would look at each of them to ensure their legitimacy. Extraction of the content of the provided researches were not used as there was no guarantee that the AI would give correct results. Tasks like those are prone to hallucinations.

A.2 Searching

Another usage was the creation of keywords that can be used to search for papers from the online databases found in the internet. This made the curation of papers easier and faster. This, however, did not replace the manual reading of the existing papers especially their RRL section to find keywords, know more about the topic, and look for possible mentioned references.

A.3 Formatting

During the writing of this paper, AI was used to automatically convert comma separated values into tables using latex syntax. This was done for convenience since the actual data was already created by the proponents.

Whenever a reference does not contain a DOI but contains its APA citation, ChatGPT is used to translate the APA format into bibtex format. This also applies to other citation formats to bibtex.

A.4 Debugging

AI was also used in debugging error messages given by latex compiler during the compilation of the PDF format.

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